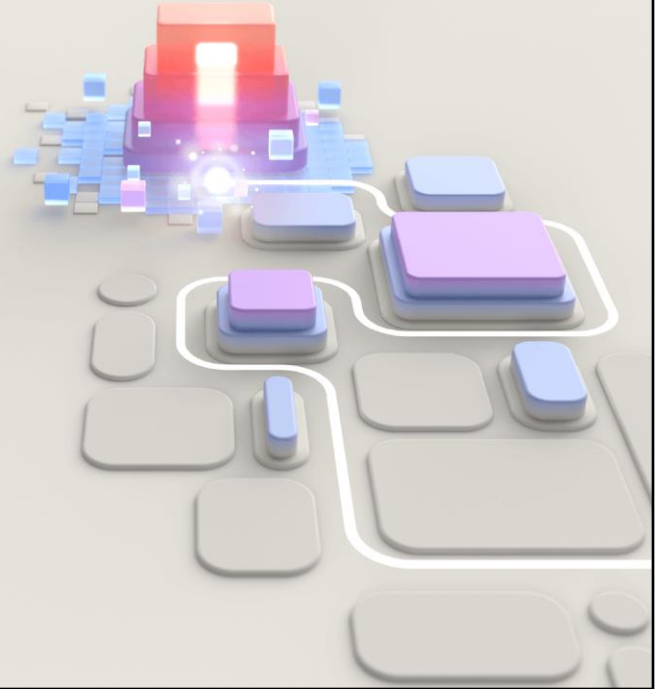




PL-400T00 Microsoft Power Platform Developer

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Automate a business process using Power Automate

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Link: <https://docs.microsoft.com/en-us/learn/modules/flow-get-started/>

Modules



- Get started with Power Automate
- Introduction to expressions in Power Automate
- Use Dataverse triggers and actions in Power Automate
- Advanced features of cloud flows

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Modules

1. Get started with Power Automate
2. Introduction to expressions in Power Automate
3. Use Dataverse triggers and actions in Power Automate
4. Advanced features of cloud flows

Module 1: Get started with Power Automate

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This module covers the following items in the JTA/OD:

- Create and manage cloud flows
- Describe flow control actions including error handling

<https://learn.microsoft.com/power-automate/triggers-introduction>

<https://learn.microsoft.com/power-automate/add-condition>

<https://learn.microsoft.com/power-automate/data-operations>

Learning Objectives

After completing this module, you will be able to:

- 1 Create a flow that automatically saves email attachments
- 2 Learn how to create a button flow to send yourself a reminder

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Microsoft Learn module

Get started with Power Automate

<https://learn.microsoft.com/training/modules/get-started-flows>



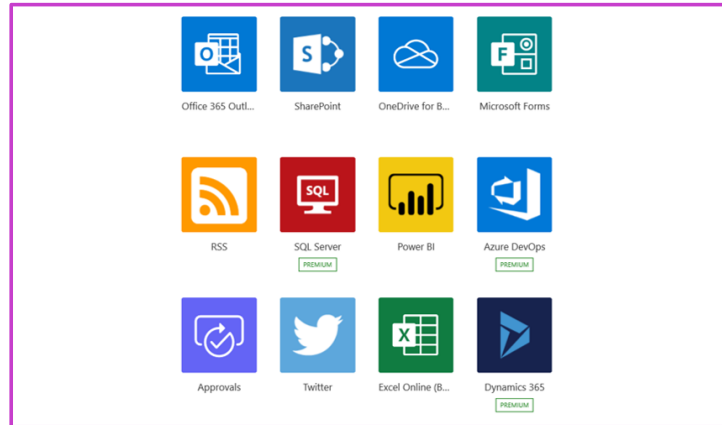
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Get started with Power Automate

<https://learn.microsoft.com/training/modules/get-started-flows>

What is Power Automate?

Power Automate is an online workflow service that automates actions across the most common apps and services

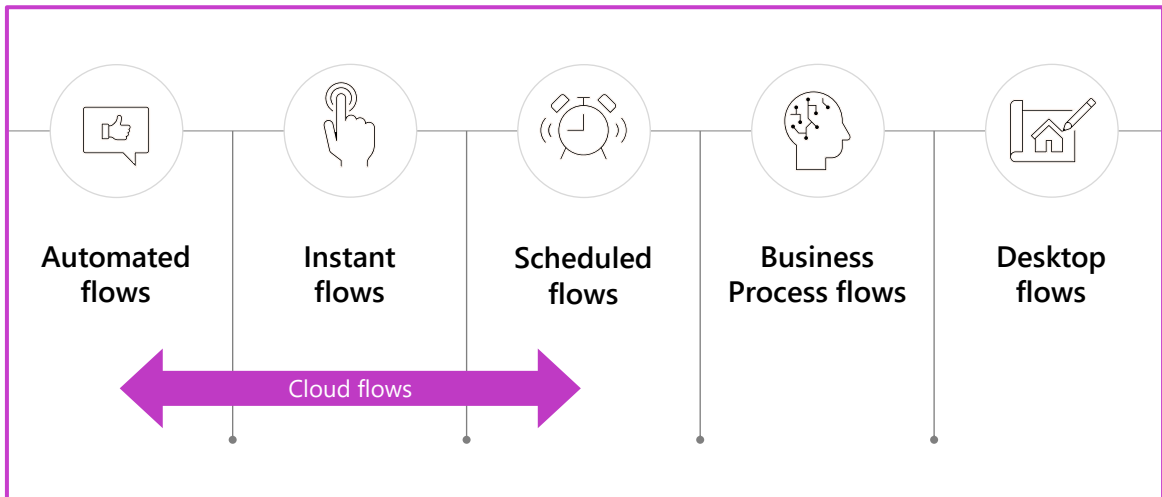


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Power Automate is an online workflow service that automates actions across the most common apps and services. For example, you can create a flow that adds a lead to Microsoft Dynamics 365 and a row in MailChimp whenever someone with more than 100 followers tweets about your company.

When you sign up, you can connect to more than 220 services, and can manage data either in the cloud or in on-premises sources like SharePoint and Microsoft SQL Server. The list of applications you can use with Power Automate grows constantly.

Types of Power Automate flows



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More on the types here <https://docs.microsoft.com/en-us/power-automate/getting-started#types-of-flows>

Automated flows – For example, on create of a project estimate row run the automation

Instant flows – For example, From the Power Automate mobile app users can click a button to run the flow on demand to start a new project estimate

Scheduled flows For example, run a weekly flow to expire old estimates

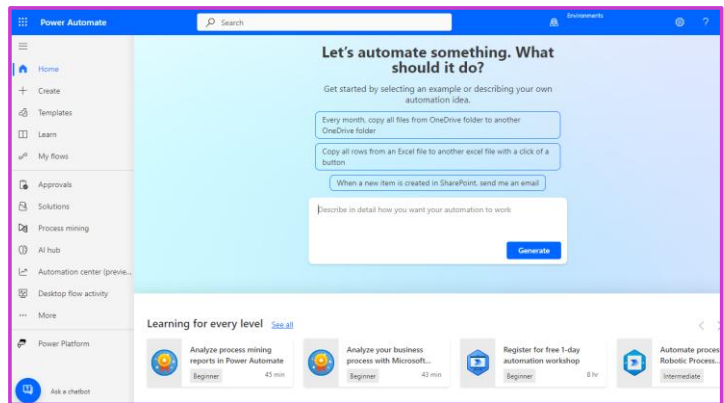
Business process flows – For example, track the stages of a project estimate

Note: Business process flows are no longer under Power Automate but are a Dataverse feature

Desktop flows –For example, automate a desktop app in the background to complete the estimate

A brief tour of Power Automate

- Flows can be created from:
make.powerautomate.com



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<https://www.microsoft.com/en-us/videooplayer/embed/RWGzz0?postIsImlMsg=true>

Let's jump into Power Automate, and we'll show you around. We have tons of information for you to learn about how to use Power Automate.

When you sign in to Power Automate, you'll find these menus:

Action items, where you can manage approvals
My flows, where your flows reside.

Create, where you start a new flow.

Templates, where you can take a look at some of the most popular templates. These should give you some great ideas for flows you want to try.

Connectors, where you can connect from one service to another.

Data, where you can access tables, connections, custom connectors and gateways.

Solutions, where you can manage your solutions.

Learn, where you can find information that will help you quickly ramp up on Power Automate.

For now, let's focus on the ? menu next to your login, which has these options:

Documentation is where our advanced topics reside. If you want to really understand a feature or function, you can do a deep dive here to figure things out.

Learn has learning paths to guide you through using Power Automate, all the way from beginning techniques to advanced scenarios.

Support is a great landing place to find help.

Roadmap is where you can get a glimpse into what will be made in the next product update.

Community is a place to plug into and find out how other people use Power Automate.

Give Feedback taps into a community of power users, and is where you can send comments and questions to developers and other experienced users.

Blog keeps you up to date about the most recent developments and releases in the Power Automate ecosystem.

Pricing can help you choose the right plan for you or your business.

Note: **Starting the flow creation from a solution will take the user to the old flow designer**
+ It is also possible some connectors are only available in old designer

Key concepts

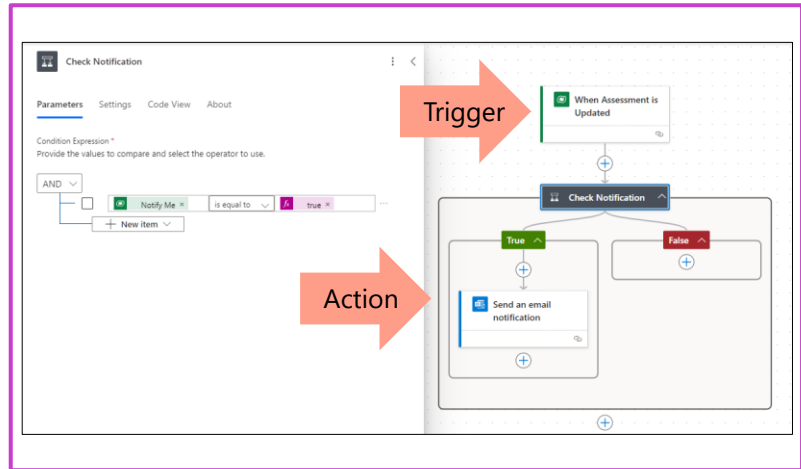
Every flow has two main parts:

A Trigger

The starting action of the flow such as new email arriving in your inbox or a new item being added to a SharePoint list

One or more Actions

What you want to happen when a trigger is invoked such as start the action of creating a new file on OneDrive for Business



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Other examples of actions include sending an email, posting a tweet, and starting an approval.

Example: Purchase order approval

The screenshot shows the 'Start and wait for an approval' configuration window in Power Automate. The left pane contains the following fields:

- Approval Type ***: A dropdown menu set to 'Approve/Reject - First to respond'.
- Title ***: A text box containing 'Please approve the Purchase order'.
- Assigned To ***: A field with a red 'X' icon and the text 'Mail'.
- Details**: A text box containing 'Markdown supported (see <https://aka.ms/approvaldetails>)'.
- Item Link**: A text box containing 'Add a link to the item to approve'.
- Item Link Description**: A text box containing 'Describe the link to the item'.

The right pane shows a flow diagram with three steps connected by plus signs in circles:

- Power Apps** (purple icon)
- Get manager** (orange icon)
- Start and wait for an approval** (blue icon)

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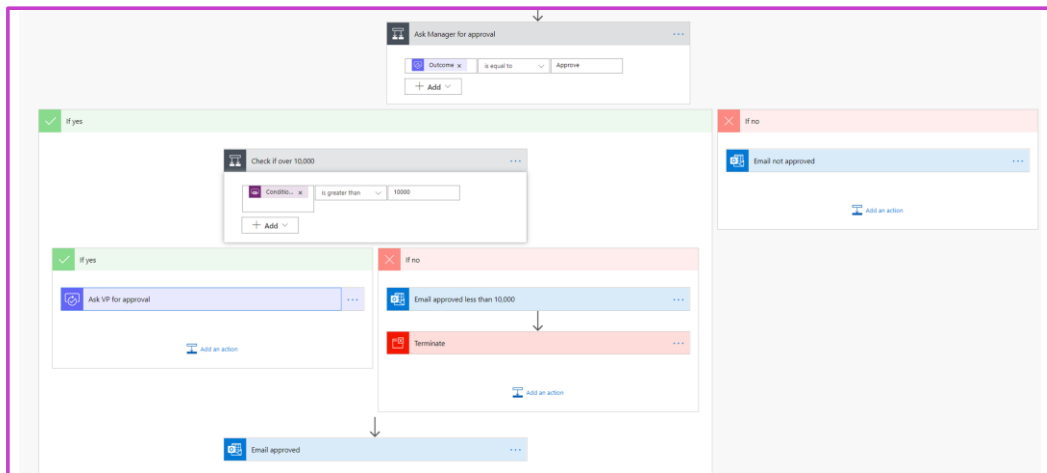
Approvals are a great process to build in Power Automate. They are often defined yet manual. Check out the following scenario for an example:

A user starts the process by going into a Power Apps app and creating a purchase order request. Once they submit the request, the information is sent to a Power Automate flow.

The flow can be built to evaluate the request and then route the request based on criteria such as submitting user and request amount. The first action could be to send the request to the user's manager. The manager could be automatically retrieved from Azure AD, avoiding prompting for duplicate information.

Here is the starting point of the flow.

Example: Purchase order approval

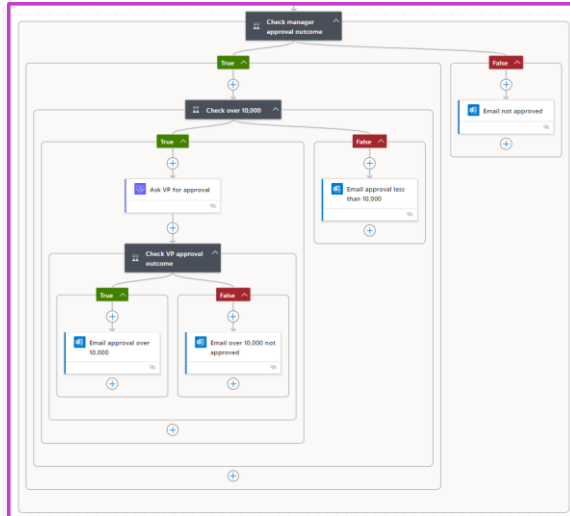


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After the manager receives the approval and approves, the flow can then provide conditional logic. Typically this might be something like if the purchase order request is greater than \$10,000 send it to VP if not then automatically approve the purchase order.

Here is an example of what this flow may look like.

Example: Purchase order approval



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After the manager receives the approval and approves, the flow can then provide conditional logic. Typically this might be something like if the purchase order request is greater than \$10,000 send it to VP if not then automatically approve the purchase order.

Here is an example of what this flow may look like.

Demo: Automated cloud flow

DEMO

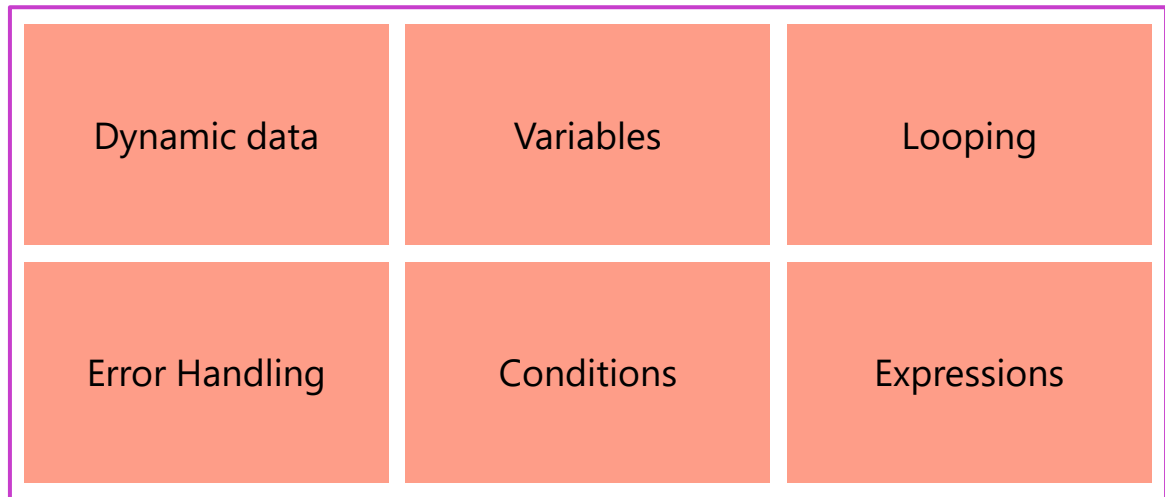
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Jump right in and build a simple cloud flow, point out each item as you build it.

When an account is created
Create a task to follow up

Build the flow interactively with the class

Flow building fundamentals

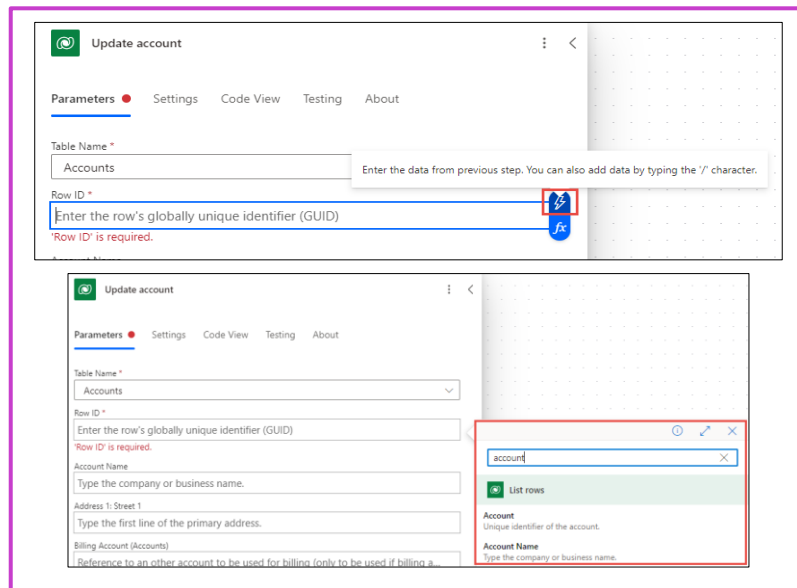


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Dynamic data

Use the Enter data from previous step and dynamically bind it to current properties

Dynamic content pane will be shown after you select the Enter data from previous step icon



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Using the dynamic content panel is how you reference data from prior steps in a flow and dynamically bind it to the current properties. The dynamic content panel is typically shown automatically when you click in a property value on a step in the flow. You can also manually open and close the panel by clicking the Add dynamic content link in the lower right-hand corner of a step.

When the panel opens it has two tabs – Dynamic content and Expression. The dynamic content tab shows a list (grouped by prior steps) of the available fields that match the data type of the currently selected field. Using this tab, you can scroll down manually and find the property you are looking for or simply search for it. Sometimes the field you are looking for doesn't show due to the data type not being exactly the same, however clicking the see more links on the step grouping often unhides those values. This allows you to use them, however, it is possible to bind to fields that wouldn't be appropriate. For example, if you bound Account Number to a date field the flow would break at runtime. You should limit your use of these additional fields to bindings you are comfortable are a good match for the data contents.

Conditions and switches

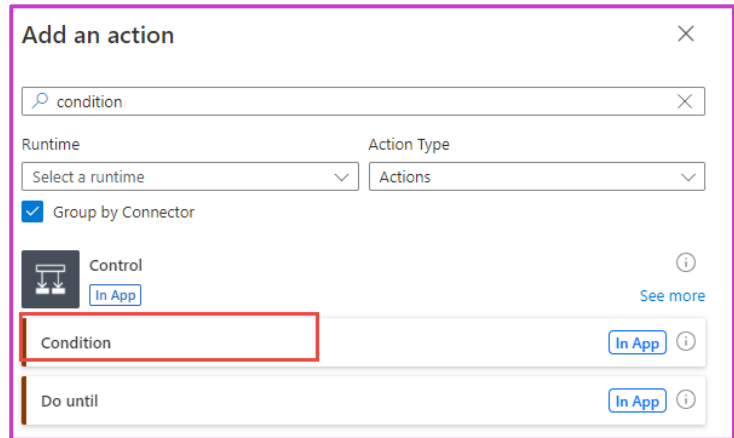
Conditions and switches offer ways to introduce conditional branching into a flow

Using conditions

Search actions on "Condition" and select

Using switches

Searching on "Switch" and add to flow



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Variables

First initialized, then set, incremented or decremented depending on their data type

Can be used in condition criteria to hold a value you plan to use later

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The image shows two screenshots of a flow tool interface, connected by a purple bracket. The top screenshot is the 'Initialize variable' action. It has tabs for 'Parameters', 'Settings', 'Code View', and 'About'. Under 'Parameters', there is a 'Name' field with the placeholder 'Enter variable name' and a 'Type' dropdown menu. The 'Type' dropdown is open, showing options: Boolean, Integer, Float, String, Object, and Array. The bottom screenshot is the 'Set variable' action. It also has tabs for 'Parameters', 'Settings', 'Code View', and 'About'. Under 'Parameters', there is a 'Name' dropdown menu with 'RecordCount' selected, and a text field with the placeholder 'Enter variable value'.

Variables in flow are like having scratch paper or a whiteboard that you jot notes on when you know you need the info later. Variables are initialized and then can be set, incremented or decremented depending on their data type. They can be used in condition criteria, used to hold a value you plan to use later, for example in an update action, or other uses where you need to hold a value for later use. To use a variable, you must first initialize it. Search actions on "var" and you will see all of the variable actions

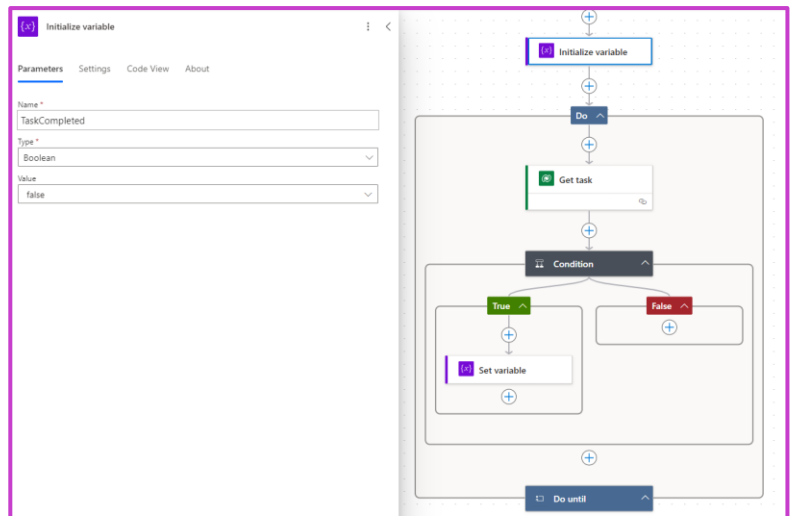
Looping

Apply for each

Executes for each item in an array

Do until

Executes until a condition is met



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Two of the constructs flow offers to do loop processing are the apply for each and do until actions. Apply for each action executes for each item in an array and do until action executes until a condition is met. Both are examples of additional flexibility Microsoft Flow provides over classic Dataverse workflows.

Using Apply for Each

The apply for each action can be used on any array, any list, of items. A common use with Dataverse is having a list records action to retrieve a set of records and then using apply for each to process the results. The following example sets the apply for each to process each of the contacts retrieved in by the ListAccountContacts action.

Using Do Until

Use the do until control when you want to loop until a condition occurs. The do until condition works similar to the standalone condition action only it is checked in a loop and controls if the loop keeps going. It's important to note that by default the loop will only occur up to 60 times, you can increase the limit to 5,000 using the change limits section.

Error handling

Goal: Recover work already completed by the flow before the error occurred

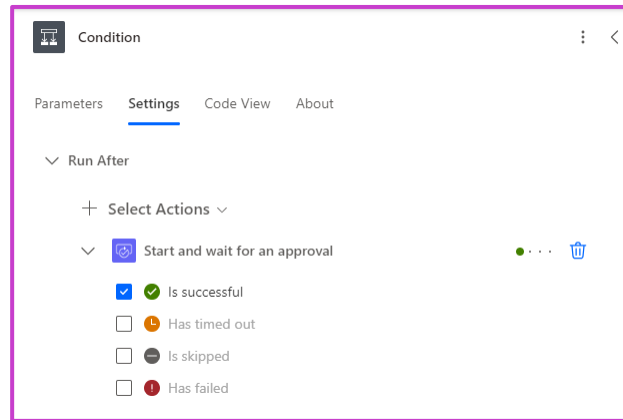
By default, flow will keep running through its step as long as prior actions have been successful

Errors:

Failed

Skipped

Timed out



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When everything goes as planned, a flow will run from top to bottom and successfully complete. An essential part of building a good flow is thinking about what can go wrong and taking the necessary steps to handle the error and compensate (recover for the work already completed by the flow before the error occurred).

For example, if during a flow's execution a record was created, then the flow did some more processing and an error occurred, you might want to clean up and remove the original record or create a task for someone to follow up and manually fix the problem. By default, when the error occurred your flow would fail and stop the execution. An administrator could resubmit the flow, but that would start execution at the top again, likely creating a duplicate record and potentially failing again. Ideally, flows would be designed to be able to be restarted with minimal manual intervention.

An important concept to keep in mind is the work completed by each action in a flow is independent from others and work completed previously will not automatically be undone if a subsequent action causes the flow to stop.

So if that is something that you need, you will then need to add the logic to execute what you would need to avoid those duplicates, or prevent a re-trigger.

Scopes can be quite useful for "grouping" related functionality, then handling errors on that group

Administering flows

You can create a flow and perform administrative tasks in a browser or, if you download the Power Automate mobile app, on your phone:

Turn flows on or off from wherever you are

See when a flow has failed

Review detailed run history reports

View and filter runs by notification type

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You can create a flow and perform administrative tasks in a browser or, if you download the Power Automate mobile app, on your phone.

Here are some of the tasks you can perform with the mobile app:

Turn flows on or off from wherever you are.

See when a flow has failed.

Review detailed run history reports.

View and filter runs by notification type.

Copilot capabilities in cloud flows

- Create a cloud flow based on a user prompt
- Edit an existing flow within the designer
- Answer questions relating to a cloud flow

Let's automate something. What should it do?

Get started by selecting an example or describing your own automation idea.

Every month, copy all files from OneDrive folder to another OneDrive folder

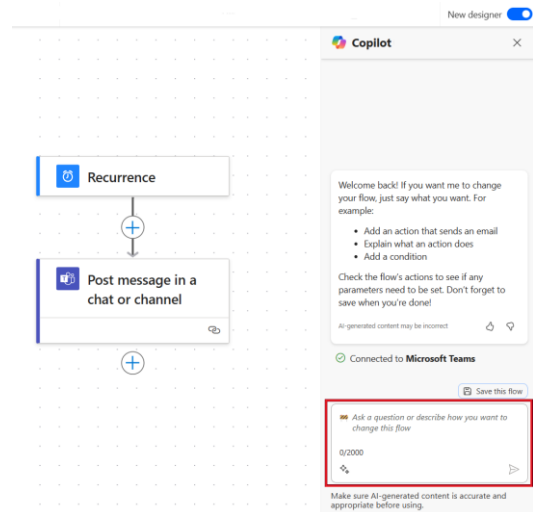
Copy all rows from an Excel file to another excel file with a click of a button

When a new item is created in SharePoint, send me an email

Describe in detail how you want your automation to work

Generate

Copilot is new technology that is still being developed. It is optimized for use with English language and has limited support with other languages. As such, parts of it may appear in English rather than your preferred language. [Learn more](#)



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Copilot in cloud flows allows you to create automation that helps streamline your workflow through quick and easy natural language expressions. You can create a flow by describing what you need through multiple steps of conversation.

The cloud flows copilot can perform the following actions:

- Understand your intent, and create a flow based on the scenario prompt you provide.
- Auto-set up connections on your behalf to get you to a working automation as soon as possible.
- Apply the necessary parameters in the flow based on your prompt.
- Respond to your requests to make changes to your flow, such as update actions and replace actions.
- Answer questions about your flow and product. For example, you can ask Copilot questions about your flow like, *What does my flow do?* You can also ask Copilot product questions like, *How do I access child flows?* and *How do I access licenses?*

Key Considerations

- Rapidly evolving area. Functionality may develop or change at pace
- May not be available in all regions; check your Dataverse environment against the Microsoft Learn
- Administrator may have disabled the functionality at the tenant / environment level
- Copilot only available in the new flow designer

Module 2: Introduction to expressions in Power Automate

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This module covers the following items in the JTA/OD:

- Implement complex expressions in flow steps

Learning Objectives

After completing this module, you will be able to:

- 1 Use one or more functions to create expressions.
- 2 Use functions to retrieve data, change data, evaluate data, and more.

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Microsoft Learn module

Introduction to expressions in Power Automate

<https://learn.microsoft.com/training/modules/introduction-expressions>

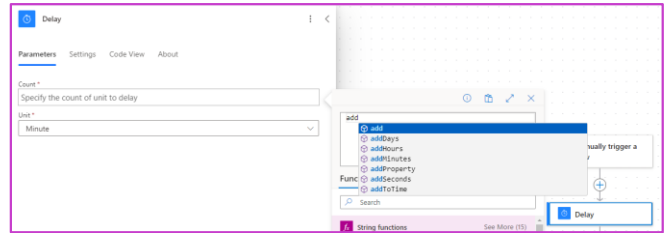


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Introduction to expressions in Power Automate

<https://learn.microsoft.com/training/modules/introduction-expressions>

Expressions



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Many reasons to use expressions:

Convert types

Perform simple, inline calculations
like string manipulation or arithmetic,
or date/time manipulation

Generate data like the current
time, a GUID or a random number

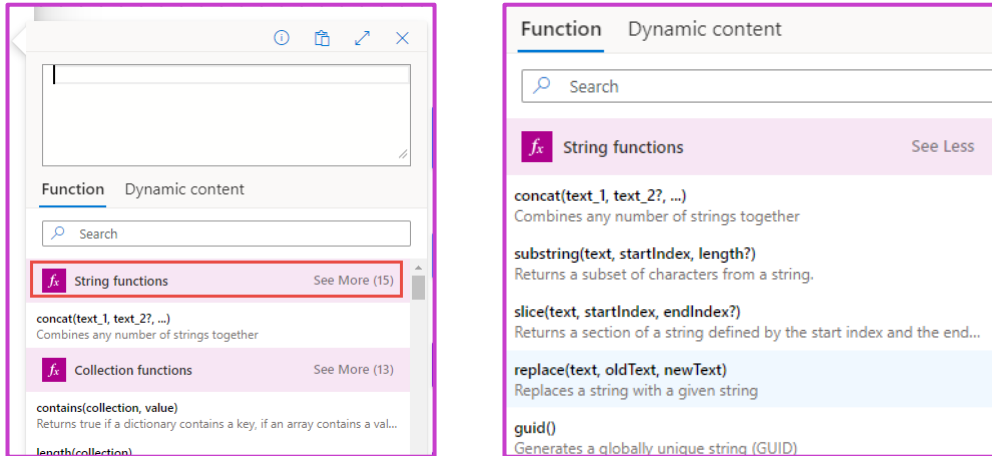
Handling optional values

"If" conditional statements / control flow

Working with lists

Types of functions

Functions are grouped into 10 different categories like math and logic.



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Functions are grouped into 10 different categories like math and logic. The categories are organized to make finding a particular function easier. Below you will get an overview of each category and some examples.

Keep in mind as you go through the various examples, we use static text and values. This is to allow you to test and recreate the examples as easily as possible. In your flows, you may substitute dynamic content in place of this static data. Just ensure that your dynamic data is the correct format for the function.

In the screenshot below, you'll notice to the right of each category header (like String functions or Collection) the words **See more**. By selecting **See more**, the complete list of functions for that category will be shown.

[!div class="mx-imgBorder"] [Screenshot of the Expression tab with the See more option for a complete list of functions.](#)

Select **See more** beside String functions.

[!div class="mx-imgBorder"] [Screenshot of See more String functions with replace\(text, oldText, newText\) selected.](#)

Module 3: Use Dataverse triggers and actions in Power Automate

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This module covers the following items in the JTA/OD:

- Configure flow steps to use Dataverse connector actions and triggers

<https://learn.microsoft.com/power-automate/dataverse/overview>

<https://learn.microsoft.com/connectors/commondataservice>

Learning Objectives

After completing this module, you will be able to:

- 1 Learn about the Dataverse triggers and actions in Power Automate.
- 2 Discover the additional inputs that are available.

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Microsoft Learn module

Use Dataverse triggers and actions in Power Automate

<https://learn.microsoft.com/training/modules/dataverse-triggers-actions>



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Use Dataverse triggers and actions in Power Automate

<https://learn.microsoft.com/training/modules/dataverse-triggers-actions>

Microsoft Dataverse Connectors



**Microsoft
Dataverse
(Legacy)**

- Use when you need to integrate across environments
- Need to use for When a row is selected



**Microsoft
Dataverse**

- Use when talking to Microsoft Dataverse in the same environment
- This is the preferred connector to use and will replace the other two connectors

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With many possible connectors, it may seem confusing. The quick answer is use the Microsoft Dataverse connector on the bottom of our list if at all possible.

The first two connectors are deprecated and should only be used for scenarios that can't be handled by Microsoft Dataverse

Microsoft Dataverse Triggers and Actions



Triggers		Actions	
Triggers Actions When a flow step is run from a business process flow PREMIUM When a row is added, modified or deleted PREMIUM When an action is performed (preview) PREMIUM Don't see what you need?	See more ① ② ③	Add a new row PREMIUM Microsoft Dataverse Delete a row PREMIUM Microsoft Dataverse Download a file or an image PREMIUM Microsoft Dataverse Get a row by ID PREMIUM Microsoft Dataverse List rows PREMIUM Microsoft Dataverse Perform a bound action PREMIUM Microsoft Dataverse Perform an unbound action PREMIUM Microsoft Dataverse	Relate rows PREMIUM Microsoft Dataverse Search rows (preview) PREMIUM Microsoft Dataverse Unrelate rows PREMIUM Microsoft Dataverse Update a row PREMIUM Microsoft Dataverse Upload a file or an image PREMIUM Microsoft Dataverse Perform a changeset request PREMIUM Microsoft Dataverse

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Read thru some of these and engage with students on what they might use any given connector for.

The Dataverse triggers are:

- **When a flow step is run from a business process flow** - This trigger allows you to start a flow when the Run Flow button in a business process flow step is clicked.
- **When a row is added, modified, or deleted** - This trigger allows you to start a flow when a row is added, modified, or deleted in a Microsoft Dataverse table, such that it matches the selected options.
- **When an action is performed (preview)** - Triggers when a Microsoft Dataverse action has completed.

The Dataverse actions are:

Add a new row - This action allows you to add a new row in the selected Microsoft Dataverse table.

Delete a row - This action allows you to delete a row from a Microsoft Dataverse table.

Download a file or an image - This action allows you to download a file or an image content from a Microsoft Dataverse table.

Get a row by ID - This action allows you to get the row that matches an

ID in a Microsoft Dataverse table.

List rows - This action allows you to list the rows in a Microsoft Dataverse table that match the selected options.

Perform a bound action - This action allows you to perform Microsoft Dataverse actions associated with a selected table.

Perform an unbound action - This action allows you to perform Microsoft Dataverse actions available in the environment that aren't associated with any table.

Relate rows - This action allows you to link a row in one Microsoft Dataverse table to another if the tables have a one-to-many or many-to-many relationship.

Search rows (preview) - This action allows you to search a Microsoft Dataverse environment using Relevance Search, and returns the rows that match the search term most closely.

Unrelate rows - This action allows you to remove the link between a row in one Microsoft Dataverse table to another if the tables have a one-to-many or many-to-many relationship.

Update a row - This action allows you to modify any selected row in a Microsoft Dataverse table, or adds a new row if it doesn't exist.

Upload a file or an image - This action allows you to upload a file or an image content to a Microsoft Dataverse table with a compatible column type.

Perform a changeset request - This action allows you to perform a group of Microsoft Dataverse connector operations as a single transaction. If one of the operations fails, all the successful actions are rolled back.

Dataverse actions in Power Automate allow you to create robust processes that surround your data.

Dataverse data change trigger



Specify for trigger

- Change type
- Table name
- Scope

When a row is added, modified or deleted

Parameters Settings Code View About

Change Type *
Added

Table Name *
Accounts

Scope *
Business unit

When a row is added, modified or deleted

Parameters Settings Code View About

Change Type *
Choose when the flow triggers

Added
Added or Deleted
Added or Modified
Added or Modified or Deleted
Deleted
Modified
Modified or Deleted

[Enter custom value](#)

Connected to admin@CRM668488.onmicrosoft.com. [Change connection](#)

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All triggers in Power Automate produce information, which is called dynamic content. Occasionally, this information only references who began the flow by selecting a button, but all Dataverse triggers give, at minimum, a row of data from a table.

When using a Dataverse trigger, you need to specify the following parameters:

Environment (Legacy connector only) - Environments are like containers in which you can organize tables, flows, apps, and more. Some organizations only use a single environment, but many have multiple environments for various priorities and developments.

Table Name - It allows you to specify the exact table where your data is held, which should trigger your flow.

Change type – Create/Update/Delete – Added/Modified/Deleted

Scope – This parameter allows you to specify when the flow runs according to the user who triggered it. Various scopes that you can choose from are:

- **Business unit** – Actions that are taken on a record that your business unit owns will trigger the flow.
- **Parent: Child business unit** – A trigger with this scope will be initiated on actions that are taken on a record that your business unit or any child business units owns.
- **Organization** – The flow will be triggered for actions that are taken on the record, regardless of owner.
- **User** – Only actions on a record that you own will trigger the flow.

One trigger allows you to request more inputs. When a user creates, updates, or deletes a record, you don't have the opportunity to ask for more information because the user could be performing those actions from multiple places. However, with the **When a record is selected** trigger, the user can trigger the flow from a button on the table inside Dataverse. Because Power Automate knows where the user is performing the action, you can request additional input for use later in the app.

Select **Add an input**, specify the type of input that you want, and then enter a short message to the user. For example, the following screenshot shows a request for an email address. The scope has not been requested because any user who has access to the table in Dataverse would be able to select the button on any record that they have access to.

Scope on triggers

Organization

Trigger for all rows user has read access to

Business unit

Only trigger for all rows user has read access to and owned by a user in same business unit

Business unit + child business unit

Only trigger for all rows user has read access to and owned by a user in same business unit and hierarchy of child business units

User

Only trigger for modifications to rows they own

When a row is added, modified or deleted

Parameters Settings Code View About

Change Type *
Modified

Table Name *
Accounts

Scope *
User

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The scope configured on the Create, Update and Delete triggers filter the triggering of the flow based on the connection user's information on the trigger action and the triggering record owner's information. The security role permissions for the trigger connection user also plays into the filtering as well. The actual owner of the Flow does not matter, it's the user that is signed in to the Dataverse connector used by the trigger that matters for filtering.

The following are the valid options for the scope setting:

Organization – This will trigger for all records the user has read access to regardless of triggering record owner's business unit.

Business Unit – This will only trigger for data records the user has read access to and owned by a user in the same business unit.

Business Unit + Child Business Unit – this will only include any of the records they have read access to in their business unit and the hierarchy of child business units.

User – This is the most restrictive it will only fire for modifications to records they are the owner of.

In Dataverse, each user and team are associated with a business unit. All data records that are user and team owned and not organization owned are associated with a business unit based on the record's user/team ownership. That business unit can either be the root (top of the chain) or a child business unit within the overall hierarchy of business units. For our examples, let's use the following structure.

Filtering Update Triggers



When a row is added, modified or deleted

Parameters Settings Code View About

Change Type *
Modified

Table Name *
Contacts

Scope *
Organization

Advanced parameters
Showing 4 of 4 Show all Clear all

Select Columns
firstname.lastname.birthdate

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<https://learn.microsoft.com/power-automate/odata-filters>

Filters and scope provide familiar concept to Microsoft Dataverse Classic Workflow features

This does not mean the value of firstname has changed, only that the column has been included in an update

Community tools like **FetchXMLBuilder** can be helpful in building the queries

Filtering Update Triggers



When a row is added, modified or deleted

Parameters Settings Code View About

Change Type *
Modified

Table Name *
Contacts

Scope *
Organization

Advanced parameters
Showing 4 of 4 Show all Clear all

Select Columns
Enter a comma-separated list of column unique names. The flow triggers i... X

Filter Rows
statuscode eq 1 X

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<https://learn.microsoft.com/power-automate/odata-filters>

The flow will only be triggered if the statuscode (status reason) is 1 (active)

You can combine column and row filters

Querying data



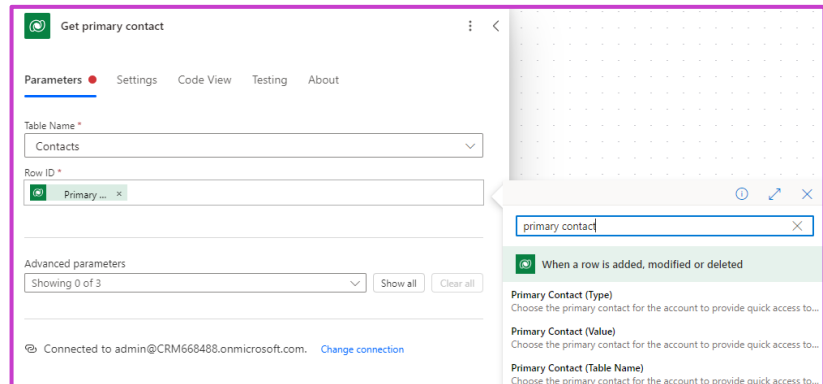
Get a row by ID action

Retrieve a single row

List rows action

Retrieve an array of rows

- OData filter
- FetchXML query



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When working with Dataverse data in a flow a common task is retrieving data. Using the Dataverse Get Record action you can retrieve a single record and using the Dataverse List Record action you can retrieve an array of records that match a filter provided. These actions are most commonly used in flows to retrieve additional data related to what caused the flow to trigger. For example, if triggered on a selected account, you might use the Get Record to retrieve a phone number for the primary contact for the account. Another scenario might be to retrieve all accounts that a contact is the primary contact of using the List Record action.

The dynamic content panel makes it easy to specify the item identifier without knowing the internal name of the lookup field which in this case would be `_primarycontact_value`. That name is important because later in this topic we will be retrieving all accounts using that field in the filter.

Remember when using either the Get Record or List Record to click on the ... and select Rename and give the action a meaningful name like `GetPrimaryContact`. This will be helpful because subsequent steps in the flow can use the data you have retrieved. For example, if you wanted to use the city on the contact you would locate city in the `GetPrimaryContact` section of the dynamic content panel list.

Creating and updating data

Address 1: Street 1
Type the first line of the primary address.

Billing Account (Accounts)
Reference to an other account to be used for billing (only to be used if billing a...

Main Phone
Type the main phone number for this account.

Primary Contact (Contacts)
fc null ×

Service Territory (Territories)
The Service Territory this account belongs to. This is used to optimize schedulin...

Function Dynamic content

Search

fc String functions See More (15)

concat(text_1, text_2, ...)

Add a new row

Provide minimum required columns to create a row

Update a row

Provide item identifier

Set column to null to clear

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The Dataverse Create Record and Update Record actions are commonly used to create and modify Dataverse data. These actions are key to automating work within the platform. Once you have selected a table, the action will list the columns that are available. They are not listed in alphabetic order so keep scrolling down if you don't see the name you are looking for. In fact, you might have to click Show Advanced to see all the columns.

When setting fields take care to ensure the data you are providing is in the correct format.

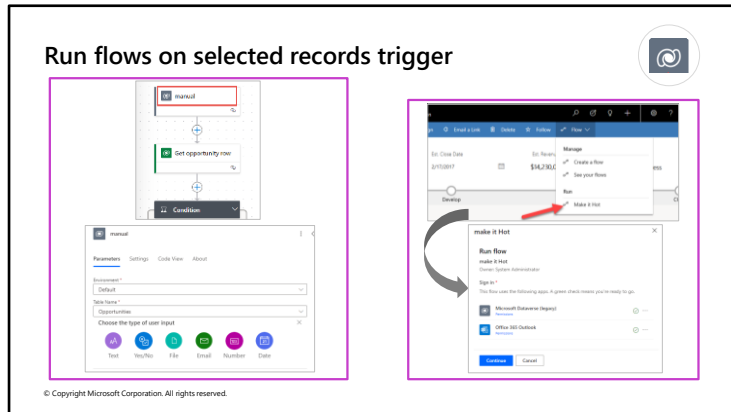
Note:

- Use null express to clear values
- Date and times are in UTC

Uses OData syntax

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This can in many cases replace what you would use Microsoft Dataverse Dialog for. since dialogs are deprecated

Note: Default for Organization Name is still used on this trigger and is the same as (Current)

Note: This requires use of the legacy connector and you should review if the new connector has been released which will make this a much more valuable feature

Module 4: Advanced features of cloud flows

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This module covers the following items in the JTA/OD:

- Manage sensitive input and output parameters
- Implement Azure Key Vault and Azure Active Directory service principals
- Add a cloud flow step to a business process flow
- Configure trigger filter and retry policies

Plus

- Utilize Power Automate cloud flows to implement business logic from a canvas app

Learning Objectives

After completing this module, you will be able to:

- 1 Handle sensitive data in flows
- 2 Manage retry policy
- 3 Differentiate flow types

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Sensitive data in cloud flow steps

Secure Inputs

Secure Outputs

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Get secret

Parameters **Settings** Code View Testing About

General

Action Timeout
Specify the duration in ISO 8601 format
Example: PTD

Networking

Content Transfer
Specify the behavior and capabilities for transferring content over HTTP

Chunk size
Specify download chunk size between and Mb. Example: 10

Retry Policy
A retry policy applies to intermittent failures, characterized as HTTP status codes 408, 429, and 5xx, in addition to any connectivity exceptions. The default is an exponential interval policy set to retry 4 times.
Default

Security

Secure Inputs
On

Secure Outputs
On

Tracking

Tracked properties

Key	Value

<https://learn.microsoft.com/power-automate/how-tos-use-sensitive-input>

Azure Key Vault

Securely store and access secrets

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Add an action



[← Return to search](#)



Azure Key Vault

Azure Key Vault is a service to securely store and access secrets.

Decrypt data with key



Encrypt data with key



Get secret



Decrypt data with key version



Encrypt data with key version



Get key metadata



Get key version metadata



Get secret metadata



Get secret version

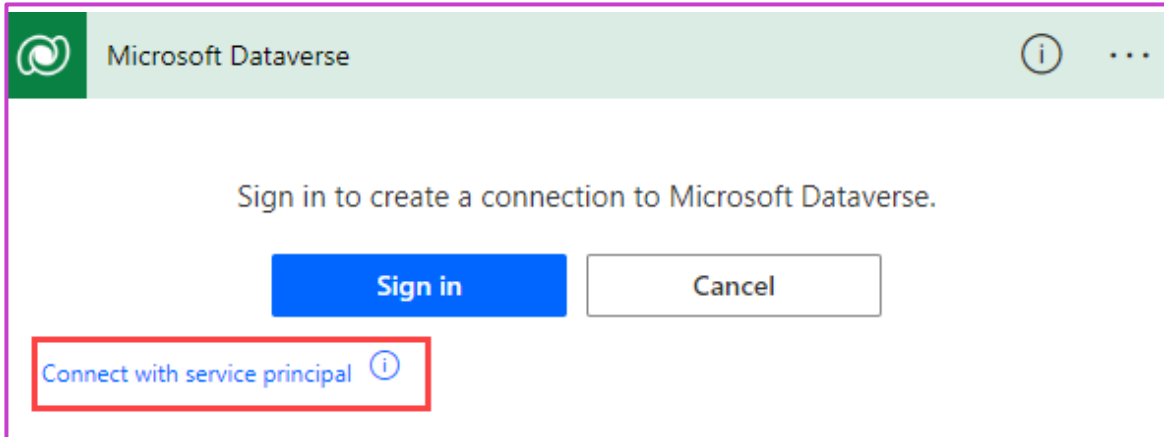


Get secret version metadata



Entra ID service principals

Dataverse connector supports Dataverse application users



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Note: Azure AD is has been renamed to Microsoft Entra Id

<https://learn.microsoft.com/azure/active-directory/develop/howto-create-service-principal-portal>

Cloud flow step in a business process flow

Business process flows

- Guides a user through business process stages in a model-driven app
- Started when a row is created
- Each BPF has a table created automatically to track current stage

Cloud flow step

- When a flow step is run from a business process flow trigger
- Instant trigger
- Flow runs on the BPF table

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<https://learn.microsoft.com/power-automate/business-process-flows-overview>
<https://learn.microsoft.com/power-automate/create-instant-flows>

Child flows

Call child flows from parent flow

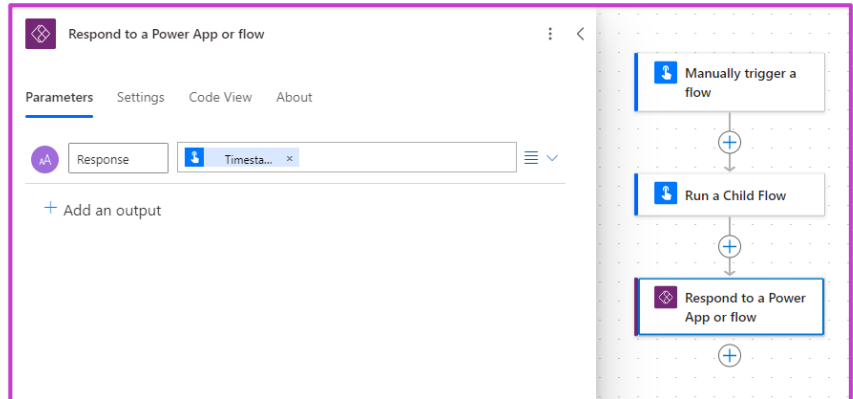
- Run a Child flow action

Child flow triggers

- Power Apps
- Manually trigger a flow

Pass data back to parent flow

- Respond to a Power Apps or flow action
- Response (from Request connector) action



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Child flows are instant flows that are called from a parent flow

Can pass parameters to a child flow and receive properties in return

The Respond to a Power Apps or flow action can only pass back simple values

The Response action, from the Request connector, can pass back JSON including multiple rows of data

<https://learn.microsoft.com/power-automate/create-child-flows>

Cloud flows and solutions

- 1 Instant flows must be created in a solution

- 2 Parent and child flows must be in the same solution

- 3 Cloud flows used in Business process flows must be in a solution

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Calling a cloud flow from a Canvas app

Call cloud flows from canvas app

- `<Flow Name>.Run(parameters)`

Flow trigger

- Power Apps
- Use Ask in Power Apps to create parameters

Pass data back to app

- Respond to a Power Apps or flow action
- Use Set to store response in a variable

```
Set(varResult,
    SendEmailFlow.Run(
        EditForm3.LastSubmit.Email,
        TextInput1.Text)
    .result
);
```

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Either create the flow from the canvas app or create a flow using the PowerApps connector

Use Ask in PowerApps to add the parameters to the flow

Use **Respond to a Power Apps or flow**

Use Set to capture response from the flow into a variable

Note: Canvas app and cloud flow must be in same solution

<https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/how-to/trigger-flow>

Learning Path 2 practice labs



Power Automate

Lab 3: Power Automate

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As we continue to build our solution, we will now create Power Automate cloud flows to process inspection data.

Summary



- Power Automate consists of a trigger and actions
- Many Connectors available including Microsoft Dataverse
- Use expressions for manipulating data
- Dataverse has many triggers and actions

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Modules

1. Get started with Power Automate
2. Introduction to expressions in Power Automate
3. Use Dataverse triggers and actions in Power Automate
4. Advanced features of cloud flows



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